occur? How can you deploy a new version of the application without downtime? With *Deployments*.

A *Deployment* is an object that manages the life cycle of a stateless, replicated application. Each replica is represented by a Pod. The replicas are distributed among the nodes of a cluster for better resilience (https://kubernetes.io/docs/reference/glossary).

In Docker you manage your application instances directly by creating and removing containers. In Kubernetes you don't manage Pods. You let a Deployment do that for you. Deployment objects have several important and valuable characteristics. You can use them to deploy your applications, roll out upgrades without downtime, roll back to a previous version in case of errors, and pause and resume upgrades.

Deployments also let you manage replication. They make use of an object named *ReplicaSet* to ensure there's always the desired number of Pods up and running in your cluster. If one of them crashes, a new one is created automatically to replace it. Furthermore, replicas are deployed across different nodes in your cluster to ensure even higher availability if one node crashes. Figure 7.4 shows the relationship between containers, Pods, ReplicaSets, and Deployments.